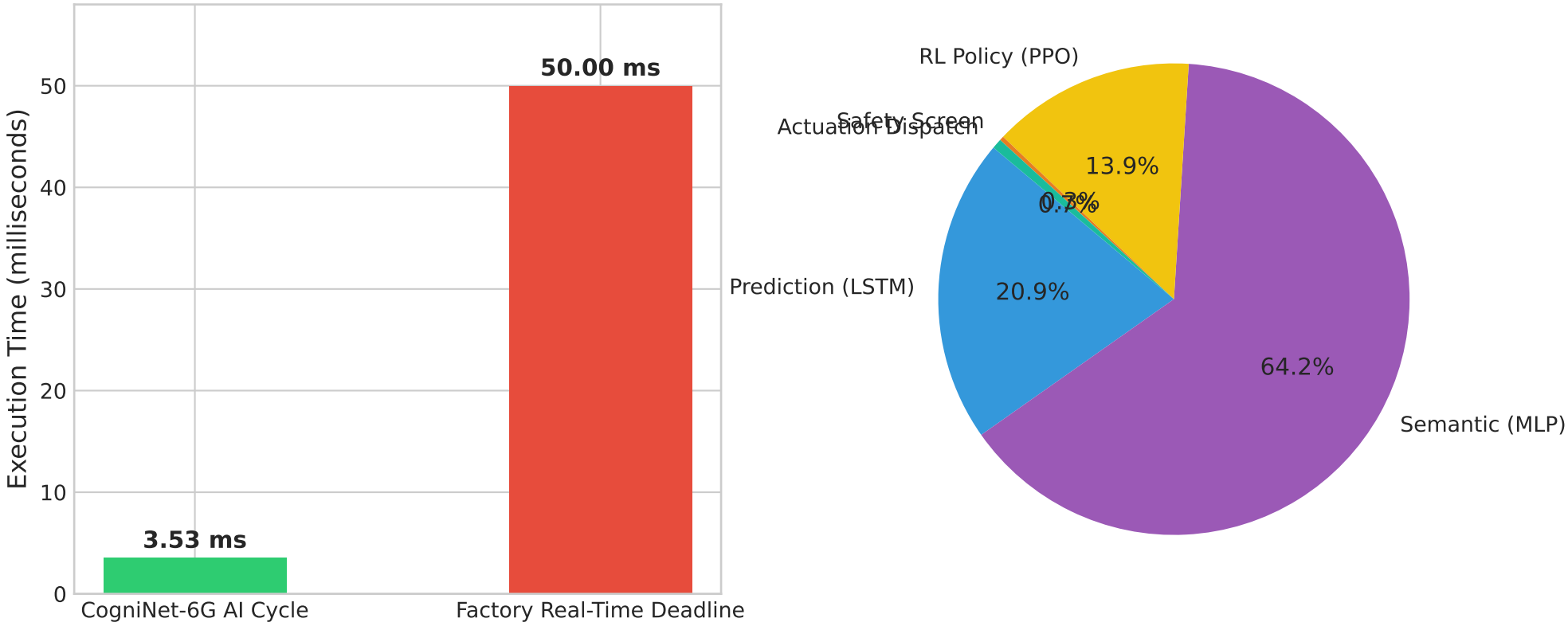
(a) Control Loop Latency vs. 50 ms Hard Deadline
Execution Time (milliseconds)
0
10
20
30
40
50
3.53 ms
50.00 ms
CogniNet-6G AI Cycle
Factory Real-Time Deadline
(b) Micro-latency Breakdown of Edge AI Components
RL Policy (PPO)
Safety Screen
Actuation Dispatch
Prediction (LSTM)
13.9%
0.7%
20.9%
64.2%
Semantic (MLP)